

# Assignment 3

August 29, 2026

Given:

Collision happening at the starting point (origin:  $t = 0, x = 0$ ). It creates two unstable particles: a neutron and an antimuon ( $\mu^+$ ).

\* Frame  $S$  is us the observers

\* Frame  $S'$  is the neutron's rest frame where it isn't moving

\* Frame  $S''$  is the antimuon's rest frame

\*  $S'$  moves at velocity  $v_n$  relative to  $S$

## 1 Part 1

For the neutron, its proper frame is  $S'$ . Since the neutron is at rest in its own proper frame, the decay event A occurs at its origin.

$$x'_A = 0$$

For the antimuon, its proper frame is  $S''$ . the antimuon is at rest in this frame meaning the decay event B occurs at its origin.

$$x''_B = 0$$

## 2 Part 2

In  $S'$ , the time elapsed between creation ( $t' = 0$ ) and decay event A is its proper lifetime:

$$t'_A = \tau_n$$

## 3 Part3

In  $S''$ , the time elapsed until decay event B is its proper lifetime:

$$t''_B = \tau_\mu$$

## 4 Part4

Using the inverse Lorentz transformation from  $S'$  to  $S$ :

$$t_A = \gamma_n(t'_A + \frac{v_n x'_A}{c^2})$$

Since  $x'_A = 0$  and  $t'_A = \tau_n$ :

$$t_A = \gamma_n \tau_n$$

## 5 Part 5

Using the inverse Lorentz transformation from  $S'$  to  $S$ :

$$x_A = \gamma_n(x'_A + v_n t'_A)$$

Since  $x'_A = 0$ :

$$x_A = \gamma_n v_n \tau_n$$

## 6 part 6

Using the inverse Lorentz transformation from  $S''$  to  $S$ :

$$t_B = \gamma_\mu \left( t''_B + \frac{v_\mu x''_B}{c^2} \right)$$

Since  $t''_B = \tau_\mu$  and  $x''_B = 0$ :

$$t_B = \gamma_\mu \tau_\mu$$

## 7 Part 7

Using the inverse Lorentz transformations from  $S''$  to  $S$ :

$$x_B = \gamma_\mu(x''_B + v_\mu t''_B)$$

since  $x''_B = 0$  and  $t_B = \tau_\mu$ ,

$$x_B = \gamma_\mu(0 + v_\mu \tau_\mu) = \gamma_\mu v_\mu \tau_\mu$$

## 8 Part 8

Using standard Lorentz transformations from  $S$  to  $S''$ :

$$t''_A = \gamma_\mu \left( t_A - \frac{v_\mu x_A}{c^2} \right)$$

substituting  $t_A = \gamma_n \tau_n$  and  $x_A = \gamma_n v_n \tau_n$ ,

$$t''_A = \gamma_\mu \left( \gamma_n \tau_n - \frac{v_\mu (\gamma_n v_n \tau_n)}{c^2} \right)$$

$$t''_A = \gamma_\mu \gamma_n \tau_n \left( 1 - \frac{v_\mu v_n}{c^2} \right)$$

## 9 Part 9

Using standard Lorentz transformations from  $S$  to  $S''$ :

$$x_A'' = \gamma_\mu(x_A - v_\mu t_A)$$

substituting  $x_A = \gamma_n v_n \tau_n$  and  $t_A = \gamma_n \tau_n$ ,

$$x_A'' = \gamma_\mu(\gamma_n v_n \tau_n - v_\mu \gamma_n \tau_n)$$

$$x_A'' = \gamma_\mu \gamma_n \tau_n (v_n - v_\mu)$$

## 10 Part 10

Using Lorentz transformations from  $S$  to  $S'$ :

$$t_B' = \gamma_n \left( t_B - \frac{v_n x_B}{c^2} \right)$$

substituting  $t_B = \gamma_\mu \tau_\mu$  and  $x_B = \gamma_\mu v_\mu \tau_\mu$ ,

$$t_B' = \gamma_n \left( \gamma_\mu \tau_\mu - \frac{v_n (\gamma_\mu v_\mu \tau_\mu)}{c^2} \right)$$

$$t_B' = \gamma_n \gamma_\mu \tau_\mu \left( 1 - \frac{v_n v_\mu}{c^2} \right)$$

## 11 Part 11

Using Lorentz transformations from  $S$  to  $S'$ :

$$x_B' = \gamma_n(x_B - v_n t_B)$$

substituting  $x_B = \gamma_\mu v_\mu \tau_\mu$  and  $t_B = \gamma_\mu \tau_\mu$ ,

$$x_B' = \gamma_n(\gamma_\mu v_\mu \tau_\mu - v_n \gamma_\mu \tau_\mu)$$

$$x_B' = \gamma_n \gamma_\mu \tau_\mu (v_\mu - v_n)$$

## 12 part 12

For the neutron to decay before the antimuon as observed in  $S$ , we require:

$$t_A < t_B$$

substituting  $t_A = \gamma_n \tau_n$  and  $t_B = \gamma_\mu \tau_\mu$ ,

$$\gamma_n \tau_n < \gamma_\mu \tau_\mu$$

expanding  $\gamma = \frac{1}{\sqrt{1 - \frac{v_n^2}{c^2}}}$ ,

$$\frac{\tau_n}{\sqrt{1 - \frac{v_n^2}{c^2}}} < \frac{\tau_\mu}{\sqrt{1 - \frac{v_\mu^2}{c^2}}}$$

squaring both sides and switching in  $v_\mu^{min}$ :

$$\frac{\tau_n^2}{1 - \frac{v_n^2}{c^2}} = \frac{\tau_\mu^2}{1 - \frac{(v_\mu^{min})^2}{c^2}}$$

rearranging:

$$1 - \frac{(v_\mu^{min})^2}{c^2} = \left(\frac{\tau_n}{\tau_\mu}\right)^2 \left(1 - \frac{v_n^2}{c^2}\right)$$

assuming the neutron is at rest in  $S'$  and setting  $v_n = 0$ ,

$$1 - \frac{(v_\mu^{min})^2}{c^2} = \left(\frac{\tau_n}{\tau_\mu}\right)^2$$

$$\frac{(v_\mu^{min})^2}{c^2} = 1 - \left(\frac{\tau_n}{\tau_\mu}\right)^2$$

$$\frac{v_\mu^{min}}{c} = \sqrt{1 - \left(\frac{\tau_n}{\tau_\mu}\right)^2}$$

since  $\tau_n \gg \tau_\mu$ ,  $\left(\frac{\tau_n}{\tau_\mu}\right)^2$  is much less than 1. applying binomial approximation ( $\sqrt{1-x} \approx 1 - \frac{1}{2}x$  for  $x \ll 1$ ): substitute  $x = \left(\frac{\tau_n}{\tau_\mu}\right)^2$ ,

$$\frac{v_\mu^{min}}{c} \approx 1 - \frac{1}{2} \left(\frac{\tau_n}{\tau_\mu}\right)^2$$

$$1 - \frac{v_\mu^{min}}{c} \approx \frac{1}{2} \left(\frac{\tau_n}{\tau_\mu}\right)^2$$

given  $\tau_n \simeq 879.4$  s and  $\tau_\mu \simeq 2.2 \times 10^{-6}$  s,

$$1 - \frac{v_\mu^{min}}{c} \simeq \frac{1}{2} \left( \frac{(2.2 \times 10^{-6})^2}{(879.4)^2} \right)$$

$$1 - \frac{v_\mu^{min}}{c} \simeq 3.13 \times 10^{-18}$$